→ Kubernetes cluster initialized and node running

```
hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ ubectl get nodes

hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl get nodes
NAME                   STATUS   ROLES           AGE    VERSION
desktop-control-plane  Ready    control-plane   6m41s  v1.34.3

hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$
```

→ Deployment created for Flask application

```yaml
  name: flask-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: flask-app
  template:
    metadata:
      labels:
        app: flask-app
    spec:
      containers:
      - name: flask-container
        image: flask-app
        ports:
        - containerPort: 5000
```

```
hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl get nodes
desktop-control-plane   Ready    control-plane   6m41s   v1.34.3

hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl apply -f deployment.yaml
deployment.apps/flask-deployment created

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$
```

→ Service configured to expose application

```yaml
apiVersion: v1
kind: Service
metadata:
  name: flask-service
spec:
  type: NodePort
  selector:
    app: flask-app
  ports:
    - protocol: TCP
      port: 5000
      targetPort: 5000
      nodePort: 30007
```

```
hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl get services

hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl port-forward deployment/flask-deployment 5000:5000
Forwarding from 127.0.0.1:5000 -> 5000
Forwarding from [::1]:5000 -> 5000
Handling connection for 5000
Handling connection for 5000
```

→ Pod running successfully in Kubernetes cluster


```
hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl get services

hp@DESKTOP-KI2M9EV MINGW64 ~/OneDrive/Documents/docker-projects (master)
$ kubectl port-forward deployment/flask-deployment 5000:5000
Forwarding from 127.0.0.1:5000 -> 5000
Forwarding from [::1]:5000 -> 5000
Handling connection for 5000
Handling connection for 5000
```


→ NodePort was not used due to Docker Desktop networking limitations, so port-forward was used to access the application locally.

localhost:5000

Docker App Running 🚀